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Sustainability requirements for connected health applications

Sofia Ouhbi[1,2] | José Luis Fernández-Alemán[1] | Ambrosio Toval[1] | José Rivera Pozo[3] | Ali Idri[4]

[1]Software Engineering Research Group, Department of Informatics and Systems, University of Murcia, Murcia, Spain
[2]TICLab Research Team, FIL, International University of Rabat, Rabat, Morocco
[3]Centro Regional de Hemodonación de Murcia, Murcia, Spain
[4]Software Project Management Research Team, ENSIAS, Mohammed V University, Rabat, Morocco

Correspondence
Sofia Ouhbi, Faculty of Informatics and Logistics, International University of Rabat, Campus de l'UIR, Parc Technopolis, Rocade de Rabat-Salé, 11100, Rabat, Morocco.
Email: sofia.ouhbi@uir.ac.ma

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Abstract

Connected health applications are efficient tools to improve health care services. The aim of this paper is to define a reusable requirements repository (catalog) for sustainable connected health applications based on the main related software engineering standards, e-health technology standards, and literature. The catalog contains requirements which cover the individual, social, environmental, and technical dimensions of sustainability. The requirements presented in this paper can be very useful to connected health application stakeholders and can be adapted to specific m-health applications. Moreover, the requirements catalog can be used to generate a checklist of requirements to evaluate and audit connected health applications. An application example is presented to show that the evaluation of connected health applications using the generated checklist is realistic.

KEYWORDS

apps, catalog, connected health, requirements, sustainability

1 | INTRODUCTION

There is a need to promote patient-self management, patient education from a very early stage, healthy lifestyle, and independence of patients.[1] All of these can be promoted through connected health (CH) applications. Connected health can be defined as "the collective term for telecare, telehealth, telemedicine, mHealth, digital health and eHealth services."[2] The CH technologies are transforming health care[3] and empowering patients as they are assuming greater responsibility for their own health care decisions.[4,5] In fact, CH can improve health care outcomes by connecting patients with caregivers and clinicians.[6] They empower people,[7,8] provide an emotional support,[9] raise awareness about chronic diseases,[10] and improve the access to supportive care.[11] The most trending CH applications are m-health applications, also known as m-health apps. M-health apps, such as apps to monitor a particular condition[12,13] and mobile personal health records (mPHRs),[14,15] which allow patients to access and coordinate their health information through their mobile devices, will play a more dominant role in CH future.[16] Many apps have been used in a variety of health areas to improve patients' lives and health.[17-19]

Sustainability. To promote a large-scale adoption of CH apps, there is a need to promote their sustainability. Achieving sustainability of health care systems is one of the objectives of research and innovation financed by the EU.[20] The definition which is the most widely used in literature[21] of sustainable development has been given by the United Nations (UN) as the ability to "meet the needs of the present without compromising the ability of future generations to satisfy their own needs."[22] According to the UN, sustainability needs to satisfy 3 dimensions: (1) economic, (2) social, and (3) environmental.[22] In addition to these dimensions, 2 other dimensions have been associated to software sustainability[21,23]: (4) individual (or human) dimension and (5) technical dimension. Software sustainability is therefore concerned with the economic, individual, social, environment, and technical dimensions which may be positively impacted by CH apps.[24] Several studies[25-27] have pointed out that requirements engineering is playing a vital role in software sustainability. Therefore, there is a need for a requirements catalog which provides different stakeholders of CH apps with different requirements set to develop and/or audit sustainable CH apps. In this paper, sustainability is defined as the extent to which an app is adopted and maintained to achieve durability and continuity in use in a given context.[28] To the best of our knowledge, no standard on sustainability has been proposed up to now. The main objective of this paper is to develop a reusable catalog for sustainable CH apps. As far as we know, there is no complete and well-defined set of requirements for sustainable CH app environments. This paper is a step forward in this direction.

Sustainable connected health catalog. The catalog presented in this paper, denominated sustainable connected health catalog (SCH-CAT), presents inherent properties of CH apps regarding sustainability. Software properties are either inherent properties or assigned properties[29] as

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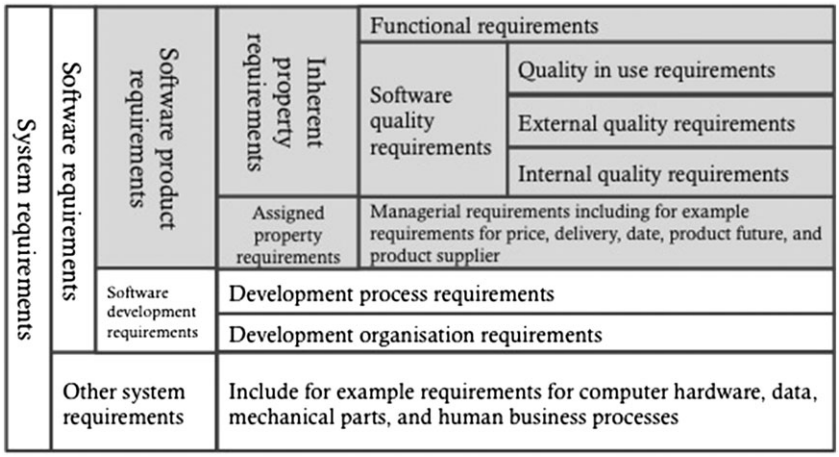

FIGURE 1 Software product requirements categorization[30]

shown in Figure 1. The inherent properties determine the capabilities of a software product.[30] They are permanent features and are opposed to assigned properties such as the software product price. The inherent properties are part of software product requirements which can be classified as either functional requirements or software quality requirements. Functional requirements determine what the software product is able to do (eg, formatting some text or modulating a signal) while quality requirements determine how well the software product performs (eg, usability requirements or performance efficiency requirements).[30] Software quality requirements help to develop sustainable software,[23] thus sustainable CH apps. These requirements are transversal to any kind of apps (PHR, prevention, treatment, and monitoring). In contrast, functional requirements are particular to a kind of apps (PHR, prevention, treatment, and monitoring). The SCH-CAT is based on the standards and recommendations from literature and is generated using requirements engineering techniques.[31]

Requirements engineering. Requirements engineering is a multidisciplinary activity of critical importance in software development.[32,33] Research in software development have found that the failures and deficiencies of software systems are often rooted in the requirements activities undertaken.[34,35] Requirements activities consist mainly of eliciting, analyzing, specifying, and validating requirements.[36,37] System requirements specification and software requirements specification (SRS) documents play a crucial role in software engineering.[38] Requirements documents are essential to both communicate requirements to stakeholders in an understandable manner and define requirements in precise detail for developers.[37]

The structure of this paper is as follows: Section 2 presents related work to this study. Section 3 presents the development of the requirements catalog for sustainable CH apps. Section 4 presents an application example of the use of catalog to evaluate CH apps. Section 5 discusses the findings of this study, implications for researchers and practitioners and limitations. Conclusion and future work are presented in Section 6.

2 | RELATED WORK

Recently, research has begun to be undertaken into how to achieve sustainable software also known as green software.[23,39,40] Green software often refers to the environmental dimension of software sustainability and can be defined as "an application that produces as little waste as possible during its development and operation."[39] Green software can be divided into[21] (1) *green in* software: produce software in a more sustainable way to achieve more sustainable product to help the environment or (2) *green by* software: develop software that are more environment friendly to support sustainability goals. In this paper, we are concerned more about *green by software* requirements. Cabot et al[41] have observed that there is a lack of standards definitions for sustainability concepts. In their study, they argue that requirements engineers should work on defining taxonomies for sustainability concepts. Many researchers[27,42] have tried to identify sustainable requirements or green requirements; however, they have focused only on requirements regarding software energy consumption. Sustainability should be defined according to the 5 aforementioned dimensions.[21]

Penzenstadler[23] explains that we should achieve a balance among the 5 aforementioned dimensions of sustainability to achieve the overall sustainability of our daily lives. Each dimension is concerned by a set of requirements.[23]

- Individual sustainability refers to the maintenance of the individual human capital, eg, health, education, skills, and access to services.[43] Individual sustainability can be covered by privacy, safety, security, human-computer interaction, usability, personal health, and well-being.[23]
- Social sustainability aims to preserve the social capital and preserve services and solidarity of social communities.[43] Social sustainability can be handled via computer-supported collaborative work[23] that aims to strengthening community building and improve community interaction.
- Economic sustainability aims to maintain capital assets and added-value (interest) assets. Economic sustainability can be taken care of in terms of costs, budget constraints, long-term business objectives, and market requirements[23] among other economic requirements.

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- Environmental sustainability seeks to improve human welfare by protecting natural resources, such as water, land, air, minerals, and ecosystem. Any system applied in a real-world context is situated within a natural environment which means that it has an impact on the environment.[23] Environmental sustainability can be managed by controlling resource flow: waste management, life cycle analysis, and environment impact assessment.[23]
- Technical sustainability refers to software systems longevity and their adequate evolution with changing surrounding conditions and respective requirements. Technical sustainability requirements include all requirements which lead to the longevity of a system such as nonobsolescence requirements and the ISO/IEC 25010[29] quality characteristics (eg, maintainability, reliability, and transferability). Moreover, energy efficiency is also part of technical sustainability requirements.[23]

There is a need for a requirements catalog that gathers sustainability requirements from software engineering standards and from literature. A requirements catalog can be applied in different scenarios and adapted to fully capture the needs and constraints of software development. Several requirements catalogs have been presented in literature for different disciplines. Toval et al[44] have presented a reusable requirements catalog to improve information systems security. Based on this catalog, Martínez et al[45] have defined a personal data protection requirements catalog to audit personal data protection. A reusable standard-based requirements catalog for learning systems development has been defined by Toval et al.[46] Reusable requirements catalog have also been presented for CH applications. Jensen et al[47] have presented reusable security requirements for health care applications. Ouhbi et al[48] have presented an audit method to assess internationalization (i18n) in e-health applications based on a reusable requirement catalog. I18n is the process of designing a software product in such a way that it can be adapted to various regions and languages without the need for engineering changes. This aspect is particularly relevant in the case of extremely sensitive information, such as that in the health domain. A recent research[49] has presented a reusable requirements catalog for i18n and sustainable but limited to blood donation apps. This paper presents a reusable requirement catalog for sustainable CH apps without a limitation to a specific context.

3 | SCH-CAT: A REUSABLE REQUIREMENTS CATALOG FOR SUSTAINABLE CH APPS

This section presents the development process of the SCH-CAT and the list of sources analyzed to select requirements for sustainable CH apps.

3.1 | Requirements specification

The content of a CH apps is very critical to its adoption by users therefore to its sustainability. Sustainable CH apps should be useful to their users; for this reason, a special attention should be taken while choosing information to be displayed to users. In addition to the content of the apps, there are other factors that influence the acceptability and adoption of CH apps, mainly software quality and i18n requirements. Therefore, requirements of i18n and software quality requirements, especially requirements related to the operability of the app, should be considered in the development/evaluation of sustainability.

Requirements for the SCH-CAT have been identified from literature reviews on CH apps[50-55] and from previous research dealing with software quality,[56-58] i18n,[48,59,60] and sustainability.[21,23,24,49] Moreover, requirements from the following standards were also extracted:

- ISO/IEC 25010 standard[29] for system and software product quality requirements and evaluation
- ISO 9241-151 standard[61] for ergonomics of human-system interaction
- ISO/TR 18307 standard[62] for interoperability and compatibility in trusted health information interchange between software apps and systems in health care
- ISO/HL7 27931 standard[63] for data exchange in health care environments
- ISO 21090 standard[64] for information interchange in health informatics
- ISO/TR 14292 standard[65] for the definition, scope, and context of personal health records
- ISO/TR 20514 standard[66] for the definition, scope, and context of electronic health records
- ISO/TS 14265 standard[67] for processing personal health information
- ISO 27799 standard[68] for information security management in health
- The W3C standards[69] for Web and mobile devices

Requirements based on Health Insurance Portability and Accountability Act of 1996 recommendations[70,71] have been included in the catalog. Furthermore, usability guidelines from app repositories[72] have been taken into consideration in addition to requirements from software engineering for sustainability project.[73] After identifying the standards and other information sources, the corresponding parts of text relevant to sustainability of CH apps have been identified, analyzed, and specified.

3.2 | Catalog development

The quality of the SRS documentation, which serves as an input to the design, coding, and testing phases, is critical to the success of any software project.[74] Therefore, for the catalog development, guidelines from the IEEE 29148 standard[38] have been followed. This standard has replaced the

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IEEE 830 standard[75] which is a guide to prepare SRS documents. Recommendations from simple reuse of software requirements[44] to develop reusable catalogs of requirements have also been followed. Simple reuse of software requirements, which is developed by the Software Engineering research Group at the University of Murcia, is a practical approach that can be used to create, select, and specify the requirements of a software system on the basis of reuse and software engineering standards. Figure 2 presents the generation process of SCH-CAT. First, we have identified main sources of requirements for CH app sustainability from standards and literature. Then, we have extracted the requirements from the selected sources. Finally, the SCH-CAT is generated. The catalog maintenance is an important task as it concerns the adaptation of requirements which are changed due to regulations, policies, and standards changes or modifications in the CH apps.[16]

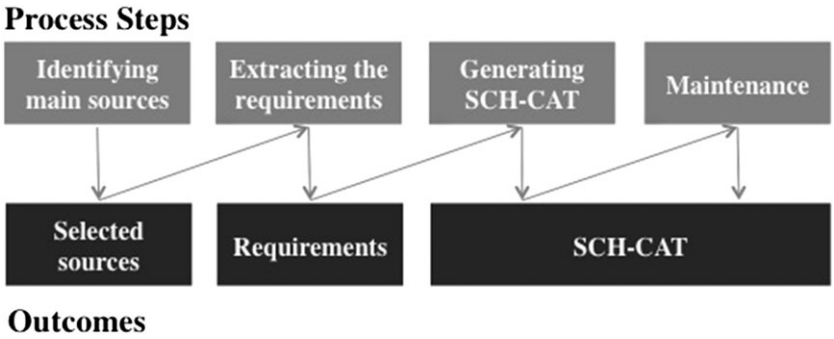

FIGURE 2 Sustainable connected health catalog (SCH-CAT) generation process

TABLE 1 Table of contents[38]

1. Introduction
1.1 Purpose
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1.3 Product overview
1.3.1 Product perspective
1.3.2 Product functions
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1.3.4 Limitations
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3.7.6.3 Environmental impact
3.7.6.4 Technical impact
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5.1 Assumptions and dependencies
5.2 Acronyms and abbreviations

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TABLE 2 Requirements for individual impact

3.7.6.1 Individual impact
1) The app shall have a positive individual impact
1.1) The app shall respect security and privacy of the user
1.1.1) The app shall have an authentication procedure
1.1.1.1) The user should be able to choose in the app's settings whether to be authenticated or not
1.1.1.2) The user should be able to access the app using a login or social network account
1.1.2) The app shall allow the user to manage personal information, such as different countries' specific legislation or regulations regarding user ownership of personal information
1.1.2.1) The user should be able to edit personal data in the profile
1.1.2.2) The user should be able to remove personal data from the profile
1.1.2.3) The user should be able to add personal data to the profile
1.1.2.4) The user should be able to share personal data from the profile
1.2) The app shall promote personal health and well-being of the user
1.2.1) The user should be able to consult health recommendations
1.2.2) The user should be able to manage his/her personal health record
1.2.2.1) The user should be able to edit the health record data
1.2.2.2) The user should be able to remove data from the health record
1.2.2.3) The user should be able to add data to the health record
1.2.2.4) The user should be able to share data from the health record
1.2.2.5) The user should be able to import data to the health record
1.2.2.6) The user should be able to export data from the health record
1.2.2.7) The user should be able to back up data from the health record

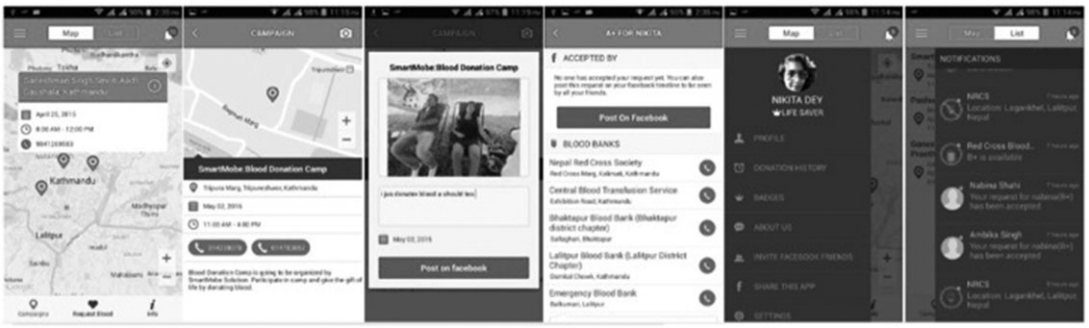

FIGURE 3 *Blood Donor +* app

3.3 | The SCH-CAT

The general structure of the requirements catalog, as recommended in IEEE 29148 standard,[38] is shown in Table 1. To facilitate the search for and reuse of sustainability requirements, along with their integration into current instantiated requirements documents for CH app that is already under way, a new subsection dedicated for sustainability requirements (ie, Section 3.7.6) was added as shown in Table 1. In the sustainability subsection, we have organized the requirements for sustainability in 4 subsections: (1) individual impact, (2) social impact, (3) environmental impact, and (4) technical impact. It should be noted that the requirements related to the economic dimension of CH app sustainability are not included in the SCH-CAT because they are not inherent requirements.

Requirements for each subsection are shown in Tables 2 to 5. Different levels of requirements refinement can be achieved to specify general requirements. For example, in Table 5, the requirement "1.2) The app shall be energy-efficient," is refined into "1.2.1) The app shall shut down in idle mode" and "1.2.2) The app shall create different profiles in the same device" which are more specific requirements. Each requirement identified has a predefined set of attributes that provides additional information on it, such as unique identifier, source, and priority. Traceability relationships are used to define relationships between the requirements identified which facilitates both reuse and maintenance tasks. As much as possible, we have followed the principles of minimal coupling and maximal cohesion to reduce the degree of interdependency (low coupling) and increase the existing conceptual and functional relationships (high cohesion) among requirements. However, some requirements are found to be associated to more than one sustainability dimension, such as the requirement concerning social networks, it is considered for social impact and for environment impact.

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TABLE 3 Requirements for social impact

3.7.6.2 Social impact
1) The app shall have a positive social impact
1.1) The app shall promote interaction among users via social networks and/or online healthcare forums
1.2) The app shall promote social solidarity among users via a rewards system
1.3) The app shall be designed for cultural diversity and multilingual use
1.3.1) The app shall adapt its content to the user's language preferences
1.3.2) The app shall be available in different languages, the languages supported and the links for selecting them should be clearly presented
1.3.3) The app shall allow the user to switch between languages at more than one point while using it
1.3.4) The app shall allow the user to choose the language of the video or audio clips
1.3.5) The app shall adapt the online help section to the user's language preferences
1.3.6) The app shall adapt the human anatomy terminology to user's language preferences
1.3.7) The app shall show the text with a correct text align depending on the user's language preferences (e.g. right align for Arabic language)
1.3.8) The app shall use appropriate formats, units of measurement or currency for international audience
1.3.9) The app shall control the advertisements showed to the user to avoid cultural discrepancies

TABLE 4 Requirements for environmental impact

3.7.6.3 Environmental impact
1) The app shall have a positive environmental impact
1.1) The app shall reduce transportation means
1.1.1) The user should be able to find nearby healthcare centers
1.1.2) If the app connects with EHRs, PHRs or third parties, it shall provide users with the possibility to interchange data
1.1.2.1) The user should be able to send data from the app to EHRs, PHRs or other parties
1.1.2.2) The user shall be able to receive data from EHRs, PHRs or other parties
1.2) The app shall be convenient for frequent use
1.2.1) The app shall use pictures to explain ideas
1.2.2) The app shall use icons to explain its contents
1.2.3) The app shall use graphs to illustrate health data history
1.3) The app shall connect to other IT resources
1.3.1) The app shall back up data in data repositories, drivers or cloud systems
1.3.2) The app shall connect to social networks
1.3.3) The app shall use device features such as Bluetooth and/or near field communication (NFC) technologies
1.3.4) The app shall connect with maps repositories to display locations

TABLE 5 Requirements for technical impact

3.7.6.4 Technical impact
1) The app shall have a positive technical impact
1.1) The app shall easily adapt with future operating system (OS) changes
1.1.1) The app shall respect mobile OS development guidelines
1.1.2) The app shall be adapted to the mobile OS platform
1.2) The app shall be energy-efficient
1.2.1) The app shall shut down in idle mode
1.2.2) The app shall create different profiles in the same device
1.3) The user should be able to have an offline access to the app
1.3.1) The user should be able to access the app without Internet connexion
1.3.2) The user should be able to edit data offline
1.3.3) The app shall save edited data offline

4 | APPLICATION EXAMPLE

This section shows how to apply the SCH-CAT to evaluate a CH app sustainability. The Android version of the *Blood Donor+* app (https://goo.gl/LOeDNS), a free CH app for blood donation, has been chosen to illustrate the evaluation process. The app *Blood Donor+* has been chosen because of its very high review score in *Google Play Store*. Some features of this app are shown in Figure 3 while Table 6 presents its characteristics.

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TABLE 6 Characteristics of *Blood Donor+* app

Android OS	4.0 and up
Price	Free
Owner	SmartMobe Solutions
Category	Social
Rating	4.7 out of 5 stars
Nbr of raters	32
Downloads	1000-5000
Update	September 2, 2016
Login	Facebook account

TABLE 7 Questionnaire regarding sustainability of the *Blood Donor+* app

ID	Question	Sustainability dimension	Req. ID	Result
Q1	Is the user able to choose in the app's settings whether to be authenticated or not?	Individual	1.1.1.1	No
Q2	Is the user able to access the app using a login or social network account?	Individual	1.1.1.2	Yes
Q3	Is the user able to edit personal data in the profile?	Individual	1.1.2.1	Yes
Q4	Is the user able able to remove personal data from the profile?	Individual	1.1.2.2	No
Q5	Is the user able to add personal data to the profile?	Individual	1.1.2.3	No
Q6	Is the user able to share personal data from the profile?	Individual	1.1.2.4	No
Q7	Is the user able to consult blood donation recommendations?	Individual	1.2.1	Yes
Q8	Is the user able to edit the blood donation record?	Individual	1.2.2.1	Yes
Q9	Is the user able to remove data from the blood donation record?	Individual	1.2.2.2	Yes
Q10	Is the user able to add data to the blood donation record?	Individual	1.2.2.3	Partially
Q11	Is the user able to share data from the blood donation record?	Individual	1.2.2.4	Yes
Q12	Is the user able to import data to the blood donation record?	Individual	1.2.2.5	No
Q13	Is the user able to export data from the blood donation record?	Individual	1.2.2.6	No
Q14	Is the user able to back up data from the blood donation record?	Individual	1.2.2.7	No
Q15	Does the app promote interaction among users via social networks and/or online healthcare forums?	Social	1.1	Yes
Q16	Does the app promote social solidarity among users via a rewards system?	Social	1.2	Yes
Q17	Does the app adapt its content to the user's language preferences?	Social	1.3.1	No
Q18	Is the app presents in its description the number of languages supported?	Social	1.3.2	No
Q19	Is the user able to find nearby healthcare centers?	Environmental	1.1.1	Partially
Q20	Does the app use pictures to explain ideas?	Environmental	1.2.1	Yes
Q21	Does the app use icons to explain its contents?	Environmental	1.2.2	Yes
Q22	Does the app use graphs to illustrate health data history?	Environmental	1.2.3	No
Q23	Does the app connect to social networks?	Environmental	1.3.2	Yes
Q24	Does the app connect with maps repositories to display locations?	Environmental	1.3.4	Yes
Q25	Does the app shut down in idle mode?	Technical	1.2.1	No
Q26	Is the app support different profiles in the same device?	Technical	1.2.2	No
Q27	Is the user able to access the app without Internet connexion?	Technical	1.3	No

The first step of the evaluation consisted of carrying out a preliminary analysis of the app's work environment and its main features and functionality. The first author then carried out the evaluation on the 14th of January 2017. A Checklist regarding the sustainability characteristics of the app was generated from the SCH-CAT. The Checklist was in the form of a questionnaire containing 27 questions so as to facilitate the work of the evaluator. We have discarded requirements from SCH-CAT that are not applicable to this app such as requirements for social impact 1.3.3-1.3.8 as the app does not adapt its content to the user's language preferences. The questions in the questionnaire can be answered by "Yes", "No" or "Partially". Table 7 presents the results of the evaluation.

The results of Q10 and Q19 were "Partially" as for Q10 donors cannot add blood donation entries manually; they have to take a picture and link it to a blood donation event. While for Q19, the app displays only nearby location in Nepal. In the description of the app, there was no mention that this app is dedicated to Nepal blood donation users. The coverage score of the *Blood Donor+* app is

Total questions: 27. Yes (+1): 12. No (+0): 13. Partially (+0.5): 2.

$Coverage = 13 * 100/27 = 48.15\%$

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The app does not achieve a moderate sustainability level (at least 50%). However, if we calculate the coverage of the app by sustainability dimension we deduce that this app has a very positive environmental impact and a moderate social impact.

$Coverage_individual = 6.5 * 100/14 = 46.43\%$,

$Coverage_social = 2 * 100/4 = 50\%$,

$Coverage_environment = 4,5 * 100/6 = 75\%$,

$Coverage_technical = 0 * 100/3 = 0\%$.

This app should address its individual, social, and technical impacts to improve its overall sustainability. A list of recommendation is provided below to improve this app's sustainability.

- To improve the app's individual impact
  - The user should be provided the option to manage his/her blood donation history and personal information manually.
  - The user should be able to choose other authentication options than login via Facebook.
- To improve the app's social impact
  - An exhaustive description of the characteristics of this app including number of languages supported and geographical limitation should be available in Google Play. Potential users can therefore check if this app answers their needs.
  - The application should be available in more than 1 language to target as many donors as possible from different backgrounds.
- To improve the app's technical impact:
  - The app should be energy efficient by turning off on idle mode
  - The app should allow users to create different accounts in the same device.
  - The app should also allow users to have offline access to its content.

5 | DISCUSSION

CH provides the potential for: enhanced reach at a relatively low cost; scalability; time efficiency; and the capacity to provide individual patients including traditionally underserved populations, patients with chronic conditions and consumers all over the world with tailoring and customization.[76] Despite the aforementioned benefits, CH apps are facing implementation, adoption and utilization challenges in practice.[77,78] We believe developing CH through the sustainability lens will enable providers to deal with these challenges. Sustainability requirements should be considered at early stages in the software development phase. However, software engineers treat sustainability as a postscript as they are driven by time-to-market pressure and do not have time to apply sustainability guidelines.[79] For this reason, the SCH-CAT can be very beneficial to facilitate the work of CH developers which will lead to the development of more sustainable CH apps. The SCH-CAT can also be very beneficial for different CH app stakeholders. In fact, developers can use all or some requirements of this catalog to develop sustainable CH apps. Health organizations or development companies which are willing to promote CH via apps can elicit, specify and validate their requirements from the SCH-CAT. Moreover, the sustainable requirements are adaptable and can be reused for specific CH apps such as apps for cardiology or oncology.

Furthermore, audit organizations or CH app stakeholders can use the SCH-CAT to evaluate and/or audit CH apps. A checklist can be generated from the SCH-CAT. This checklist should include a set of requirements that will constitute the list of items to be checked and can be adapted to the CH app to be evaluated. To that end, a reuse-based approach to generate new requirements documents from the SCH-CAT should be applied as shown in the application example. This step is necessary, since the variation points of the reusable requirements, which are basically parameterized requirements and traceability relationships, must be resolved in reuse time in accordance with the system evaluation and/or audit. The requirements documents will then be mapped onto items of the Checklist in order to carry out the evaluation and/or the audit of CH apps.

This study may have several limitations, such as other sources relevant to the generation catalog may have not been taking into consideration. However, the fact that requirements were elicited from several sources in software engineering, e-health technology and literature about sustainability may have alleviated this issue. Also, using only one app in the application example may not be sufficient to demonstrate the applicability of the catalog. However, the application example was presented to give insights on how to use the SCH-CAT to generate a checklist that can be used in the evaluation of CH apps. The SCH-CAT has certain contextual bounding (e.g. suggesting social media as a login), the findings must be therefore interpreted in light of these contexts. Requirements about the economic dimension of sustainability have been omitted from the SCH-CAT, therefore, the catalog does not cover all aspects of sustainability. However, such requirements are linked to the assigned properties of the CH software product and they are subject to change without including changes in the CH app's source code.

6 | CONCLUSION AND FUTURE WORK

In spite of the increasing interest in sustainability in our digital-driven globalized world, no specific standards dealing with sustainability were found. The requirements presented in this study have been primarily established from the results of literature reviews conducted in this domain and anal-

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ysis of standards related to software engineering and e-health. The SCH-CAT can be applied in different scenarios and adapted to fully capture the needs and constraints of CH app development. The practical applicability of the SCH-CAT has been shown by means of an application example. It has been demonstrated that the app used in the application example has to improve its individual, social and technical impacts to improve its overall sustainability. This work can be useful to requirements engineering researchers and practitioners concerned with sustainability of CH apps.

For future work, we intend to conduct an independent industry-led evaluation of the catalog. We intend also to make progress in the improvement of the requirements catalog by harmonizing the multiple requirements that are scattered throughout different sources.[80,81] The requirements catalog contents will be subject to continuous improvement through the incorporation of new knowledge originating from additional information sources, such as other recommendations, standards or policies. Moreover, we intend to advance the knowledge about software sustainability in literature.

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ORCID

Sofia Ouhbi http://orcid.org/0000-0001-7614-9731

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